Bankruptcy Prediction System

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Abstract

Early detection of financially distressed companies is critical for mitigating financial risk. Traditional predictive models often fail in imbalanced datasets, where bankrupt companies are a minority. This study develops a recall-optimized bankruptcy prediction system using Logistic Regression, Random Forest, XGBoost, and Neural Networks. Key contributions include prioritizing recall, implementing cost-sensitive learning and threshold tuning, and evaluating models on recall, precision, F1 score, and AUC. XGBoost demonstrated the best balance between high recall and acceptable false positives, making it suitable for practical deployment.

1 Introduction

Detecting bankrupt companies early is vital for financial institutions to avoid losses, regulatory penalties, and reputational damage. Imbalanced datasets make this task challenging, as the minority class (bankrupt companies) is easily overlooked. This study focuses on building a machine learning system that maximizes **recall**, ensuring the model identifies as many bankrupt companies as possible.

2 Problem Definition

- **Objective:** Identify companies likely to go bankrupt with high sensitivity.
- **Target Variable:** `Bankrupt?` (1 = bankrupt, 0 = non-bankrupt)
- **Challenge:** Highly imbalanced dataset (minority class <0.48%).
- **Key Metric:** Recall (sensitivity) to minimize missed bankruptcies.

3 Dataset Description

Property	Details
Source	Taiwan Economic Journal for the years 1999 to 2009
Number of rows	6819
Number of features	95
Target distribution	Minority class $< 0.48\%$

Table 1: Overview of the Dataset

3.1 Preprocessing Steps

1. Remove low-variance features.
2. Drop rows with missing values.
3. Standardize features using `StandardScaler`.
4. Split the dataset into training (70%), validation (15%), and test (15%) sets using stratified sampling.

4 Methodology

4.1 Handling Class Imbalance

Class imbalance was addressed using cost-sensitive learning:

Model	Cost-Sensitive Strategy
Logistic Regression	`class_weight={0:1,1:5}`
Random Forest	`class_weight='balanced'`
XGBoost	`scale_pos_weight = imbalance ratio * 1.5`
Neural Network	Class weights computed from training data

Table 2: Class Imbalance Handling Strategies

4.2 Model Selection

Four models were selected to capture a range of patterns:

Model	Type	Advantages	Limitations
Logistic Regression	Linear	Interpretable, simple	May underfit non-linear data
Random Forest	Ensemble Trees	Captures non-linear patterns, robust	Potential overfitting
XGBoost	Gradient Boosting	Handles imbalance, high discriminative power	Requires careful tuning
Neural Network	Deep Learning	Captures complex relationships	Data-intensive, overfitting risk

Table 3: Models Evaluated

4.3 Hyperparameter Tuning

Models were tuned using `GridSearchCV` with **recall** as the scoring metric:

- Logistic Regression: Regularization parameter C.
- Random Forest: Number of estimators, max depth, min samples per leaf.
- XGBoost: Max depth, learning rate, number of estimators.
- Neural Network: Early stopping and dropout regularization.

4.4 Threshold Tuning

The default 0.5 decision threshold was adjusted on the validation set to maximize recall, ensuring improved sensitivity for bankrupt companies.

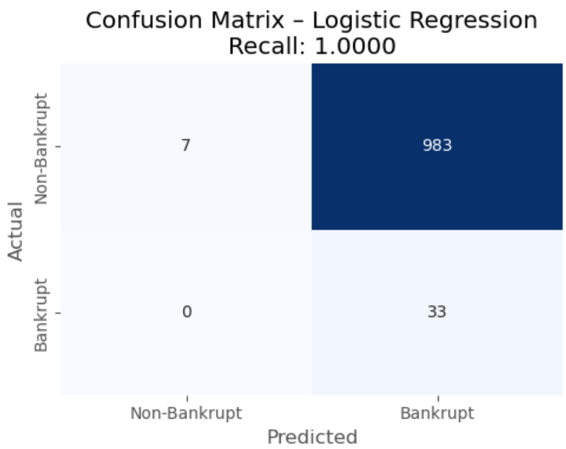
5 Results

5.1 Fine-Tuned Metrics on Test Set

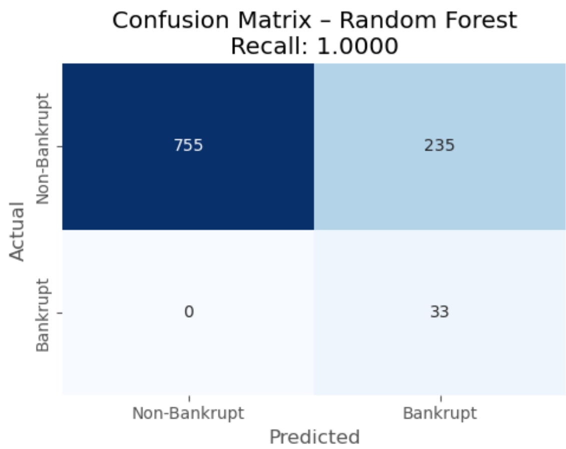
Model	Accuracy	Precision	Recall	F1	AUC	Threshold
Logistic Regression	0.0391	0.0325	1.0000	0.0629	0.8908	0.0000
Random Forest	0.7703	0.1231	1.0000	0.2193	0.9458	0.0366
Neural Network	0.0323	0.0304	0.9394	0.0589	0.8500	0.0000
XGBoost	0.8759	0.1948	0.9091	0.3209	0.9382	0.0067

Table 4: Test Set Performance Metrics with XGBoost Highlighted

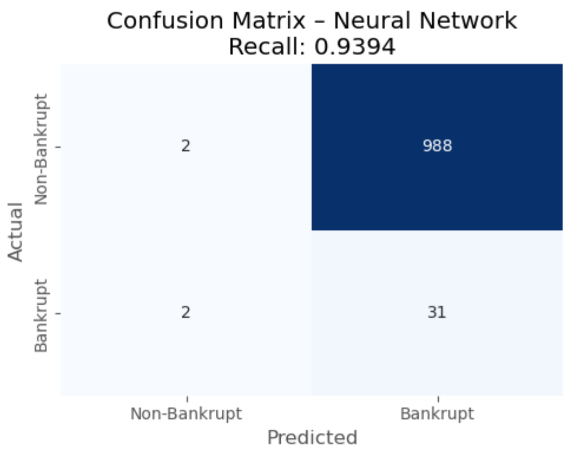
5.2 Confusion Matrices



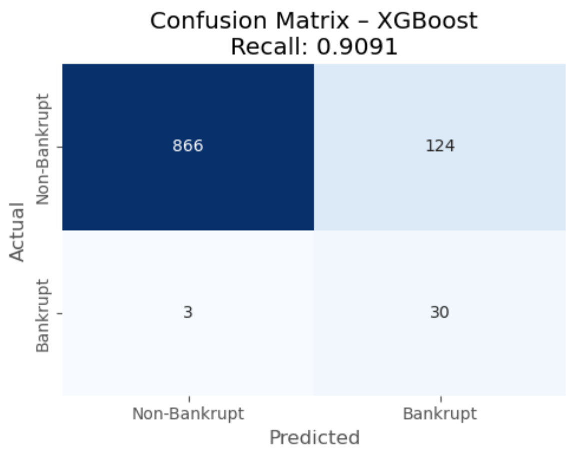
(a) Logistic Regression



(b) Random Forest



(c) Neural Network



(d) XGBoost

Figure 1: Confusion Matrices of All Models

5.3 ROC Curves

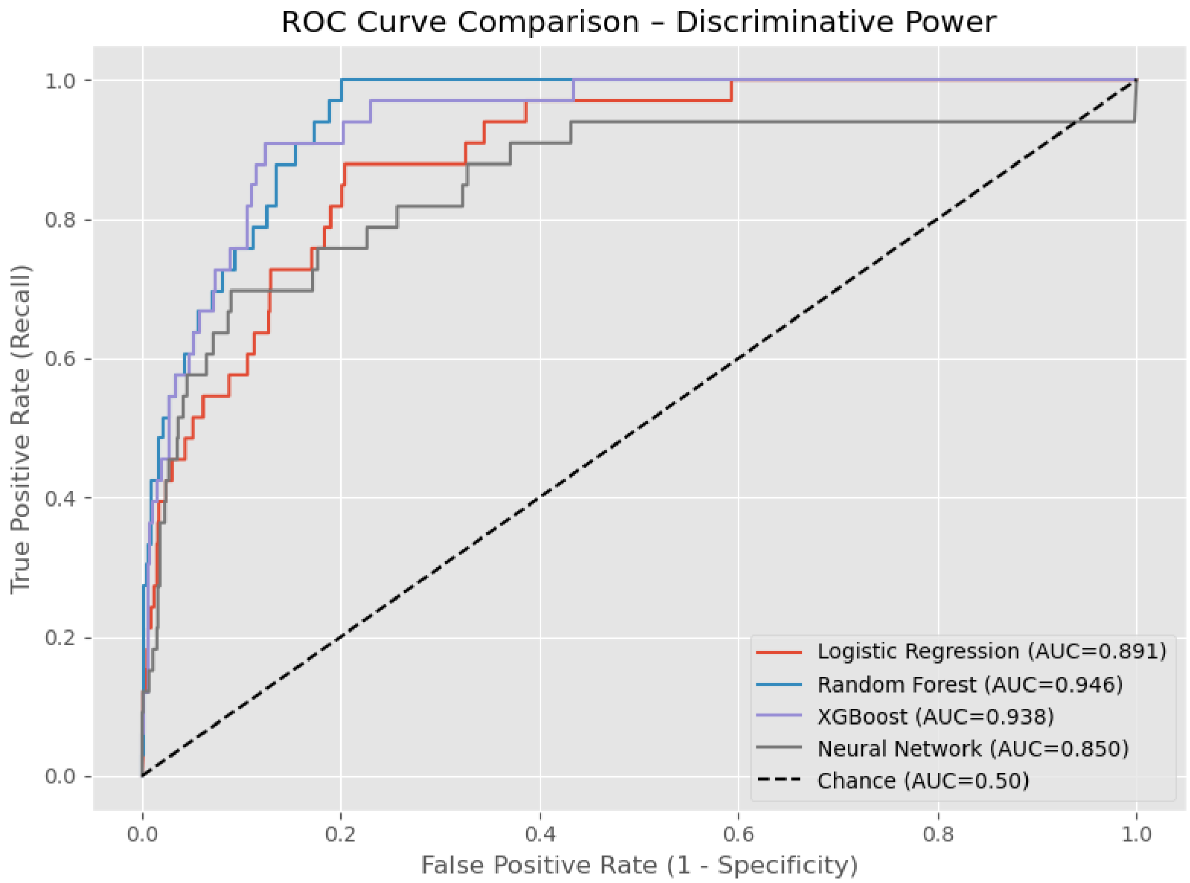


Figure 2: ROC Curve Comparison for All Models

5.4 Interpretation of Results

- Logistic Regression & Neural Network: High recall but low precision, indicating over-prediction of bankruptcies.
- Random Forest: Maintains perfect recall with moderate precision.
- XGBoost: Best balance of recall, precision, and AUC, making it suitable for real-world deployment.

6 Practical Deployment Recommendations

1. Deploy XGBoost as the primary model.
2. Use threshold tuning to adjust sensitivity according to operational priorities.
3. Continuously monitor recall, precision, F1 score, and AUC in production.
4. Retrain periodically with updated financial data.
5. Review tolerance levels for false positives and false negatives regularly.

7 Conclusion

This study highlights the importance of **recall optimization** in bankruptcy prediction. Missing bankrupt companies (false negatives) has severe consequences in financial risk management. By applying cost-sensitive learning and threshold tuning:

- Logistic Regression and Neural Networks achieved near-perfect recall but with low precision.
- Random Forest maintained perfect recall with moderate precision.
- XGBoost offered the best balance, with high recall, improved precision, and the highest AUC.

Deploying XGBoost with dynamic thresholding and regular retraining provides a robust framework for **early detection of financially distressed firms**, supporting informed decision-making and proactive risk mitigation.